

현대대수학 개론 (TP310-01)

학기	2026년 1학기
담당교수	Hudson Thomas Bryan
이수구분	교선�필수
학점	3.0
강의시간	화09:00-10:30(E7 - 224), 목09:00-10:30(E7 - 224)
강의실	E7 L12 (교수실)
소속	기초학부

교과개요

If in its classical form algebra essentially deals with the problem of finding solutions to equations, its modern formulation has at its core the notion of algebraic structure: a set endowed with certain operations, which satisfy certain requirements. Algebraic structures are ubiquitous, not only in mathematics, but in science at large, as proven by the centrality of Linear Algebra (which is essentially the study of the algebraic structure known as vector space) and Homological Algebra (used in Topological Data Analysis). In this course we will cover the basics of three of the most important algebraic structures: groups, rings and fields.

교과목표

The goal of this course is to familiarise the students with both the notion of algebraic structure as well as the techniques used to study and classify their objects. The students will be also exposed to the language of category theory, as it is a very useful for providing a common framework to many different constructions.

주차별 강의계획

1주차:Equivalence relations; Set theoretic construction of the natural, integer and rational numbers;
2주차:Algebraic structures, monoids, groups, rings and fields; $\mathbb{Z}/m\mathbb{Z}$, symmetric group;
3주차:Homomorphisms, (normal) subgroups, kernel, image; Isomorphism theorems;
4주차:Cyclic groups, Quotient Groups; Introduction to category theory;
5주차:Universal constructions; Free groups, generators and relations;
6주차:Product of groups; Group actions, orbits, stabilisers;
7주차:Actions from linear algebra, $GL_n(\mathbb{R})$; Ring homomorphisms, ideals;
8주차:Midterm
9주차:Prime and maximal ideals, unique factorisation domains; Operations on ideals;
10주차:Polynomial rings; Quotient rings, isomorphism theorems;
11주차:Euclidean domains, division algorithms; Principal ideal domains;
12주차:Algebraic and trascendental field extensions; Splitting field of a polynomial;
13주차:Algebraic closure; Fundamental theorem of Algebra;
14주차:Galois extensions and Galois group; Fundamental theorem of Galois theory;
15주차:Symmetric functions and Schur polynomials; Links with algebraic combinatorics;
16주차:Final

학습윤리

Cheating and plagiarism will not be tolerated and, if discovered, will lead to failing the entire course. This also applies to students who knowingly allow their work to be copied.

교과정책(방침)

In order to pass the course students are expected to score at least 20W% on both the midterm and the final. In particular, not attending either the midterm or the final will result in a failing grade. Exceptions to this general rule will be granted on a case-by-case basis at the discretion of the instructor.

Make-up exams will be granted only in exceptional cases, following the submission of appropriate documentation explaining the circumstances.

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준비물 및 기타

Lecture notes and the recordings of the lectures will be made available on LMS.